# Critical Minerals, Geopolitics, and the Green Transition

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# Critical minerals will power the green transition

- Lithium, nickel, cobalt, and other minerals
  - Critical for producing advanced batteries
  - Concentrated geographically, but traded globally
- Concentrated endowments give countries market power

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# Indonesia Weighs Deep Cuts to Nickel Mining to Boost Prices

- Energy ministry considering big reduction, people familiar say
- Prices of the battery metal have slumped on booming supply

By Eddie Spence and Faris Mokhtar
December 19, 2024 at 6:55 AM GMT-3

BENCHMARK SOURCE
Supply Chain Intelligence for the Energy Transition

## Australian lithium supply cut by 15% due to low spodumene prices

14th November 2024

Fastmarkets

## Cobalt export quotas: DRC sets limits to rebalance global supply

**The Democratic Republic of Congo has introduced cobalt export quotas following the suspension period, setting limits on shipments while outlining future allocations. The policy is designed to balance global supply and demand while supporting the country's ambition to develop domestic processing capacity.**

**September 22, 2025**

MINING.COM

## South America looks at creating "lithium OPEC"

Cecilia Jamasmie | March 6, 2023 | 6:53 am Intelligence News Suppliers & Equipment Latin America Lithium

# This paper

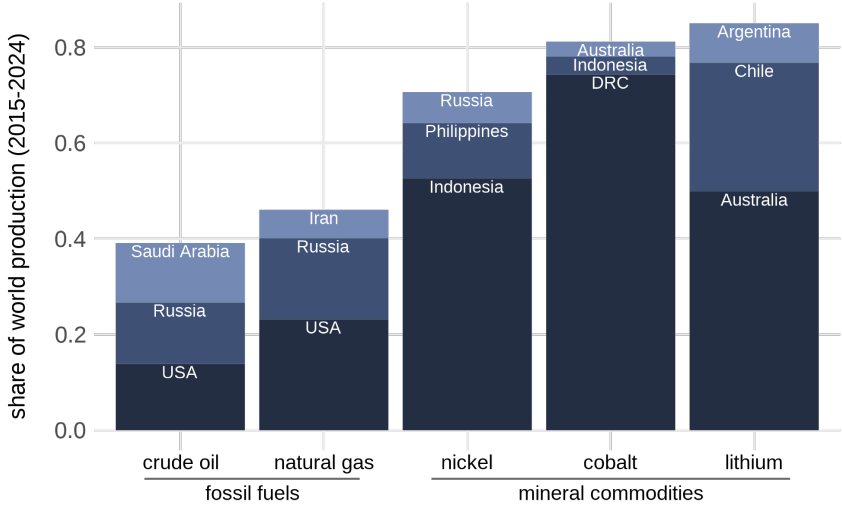
- What are the effects of energy resource concentration?
  - Market power and interdependence (geopolitics)
  - Global battery adoption (green transition)
- Minerals are joint inputs in battery production
  - We need to study these markets jointly
- Dynamic equilibrium model of global mineral markets

# Mechanisms

- Market power with complements $\neq$ market power with substitutes
- Complementarity affects interdependence
  - Saudi oil cut $\rightarrow$ US oil gains (substitutes)
  - Indonesian Ni cut $\rightarrow$ Australian Li loses (complements)
- Complementarity affects adoption
  - Within-mineral consolidation $\rightarrow$ raises markups (substitutes)
  - Across-mineral consolidation $\rightarrow$ eliminates double markups (complements)

# Background

# Mineral supply is geographically concentrated



# Critical minerals

- Minerals with low substitutability + high supply risk
  - Focus on criticality for energy transition (lithium, cobalt, nickel)
  - Separate national security purposes (rare earths)

| Batteries | % |
|---|---|
| EV | 80 |
| BESS | 15 |
| Portable | 5 |

| EV batteries | % |
|---|---|
| Lithium | 65 |
| Cobalt | 45 |
| Nickel | 11 |

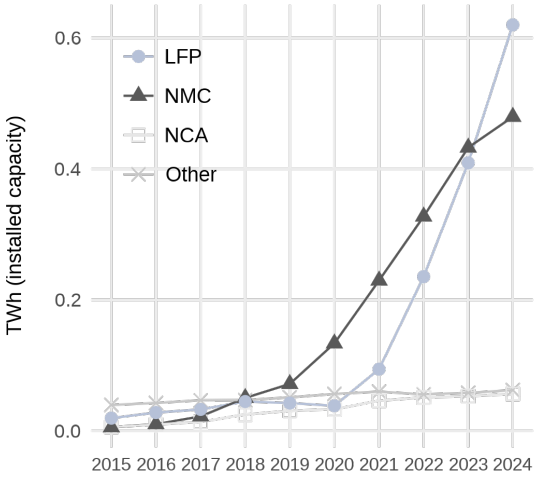
# Stages

1. Upstream mining and processing
   - Australia, Indonesia, DRC with Chinese ownership
2. Midstream refining and battery manufacturing
   - China
3. Downstream EV manufacturing
   - China, US, Europe, Korea, Japan

# Data

- Mineral supply: mine-level production and costs (GlobalData, S&P, BMI)
  - Near universe of world production for lithium, nickel, cobalt
- Battery demand: EV producer-level consumption (Rho Motion)
  - E.g., Tesla's purchases of NMC batteries for Model S production in US, Q1 2021
- Battery-mineral "recipes" (Argonne National Laboratory)

# Mineral recipes vary by battery technology

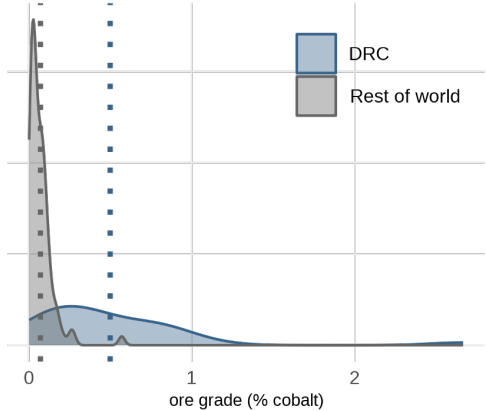



| | nickel | lithium | cobalt |
|---|---|---|---|
| LFP | - | 0.09 | - |
| NMC | 0.47 | 0.11 | 0.18 |
| NCA | 0.65 | 0.1 | 0.12 |

kg mineral/kWh capacity

# With heterogeneous costs across mines



# Framework

# Market concentration with complementarity

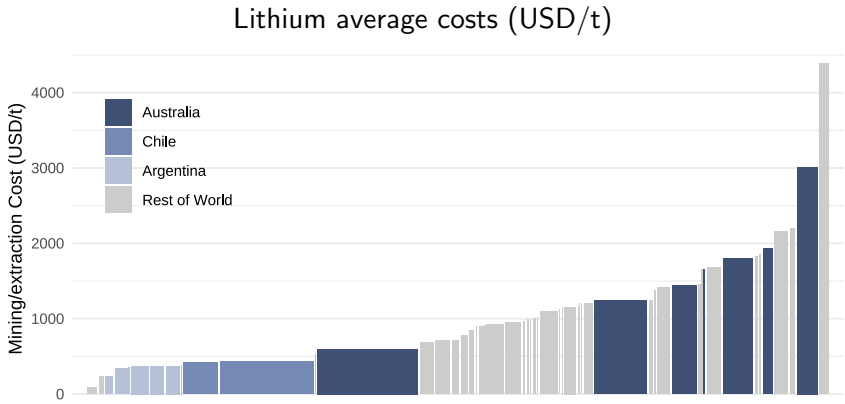
- Indonesia exercises market power for nickel
  - Nickel price ↑, Indonesian profits ↑
  - But consumers switch away from nickel-heavy batteries
- Policy spillovers for Australian lithium
  - Helped via **substitution** toward lithium-heavy batteries
  - Hurt via **complementarity** as nickel-heavy batteries still use lithium
- Green adoption in aggregate
  - Slower if **gross substitutes** (both prices ↑)
  - Ambiguous if **gross complements** (Ni price ↑, Li price ↓)

# Empirics

1. Mineral **supply** $s^m$ with dynamic extraction (geopolitics)
2. Battery **demand** $d_j$ across technologies (green transition)
3. **Recipes** $r_j^m$ connect minerals and batteries

# Supply

# We construct supply curves from our cost data



# Simplest model: myopia, $mc = ac$, price-taking

- But with myopia, no **dynamics** from finite reserves
- With $mc = ac$, no **cost convexity**
  - Local approximation only for small changes
  - Predicts all-or-nothing production and discrete jumps
- We keep price-taking by mines (with market power by country)

# Supply by mine $i$, year $t$

| Euler condition | Marginal revenue | Marginal cost |
|---|---|---|
| $mr^{it} - mc^{it} = \beta \mathbb{E}^{it}[mr^{it+1} - mc^{it+1}]$ | $mr^{it} = p^t g^i$ | $mc^{it} = \kappa x^{it} + \varepsilon^{it}$ |

- **Dynamics** from finite reserves (Hotelling)
  - Extraction $x^{it}$, price $p^t$, grade $g^i$ (output $g^i x^{it}$)
- **Cost convexity** $\kappa$, which we estimate
  - Cost shocks $\varepsilon^{it}$

# Estimation

$$\underbrace{p^t g^i}_{mr^{it}} = \underbrace{\kappa x^{it} + \varepsilon^{it}}_{mc^{it}} + \eta^{it}$$

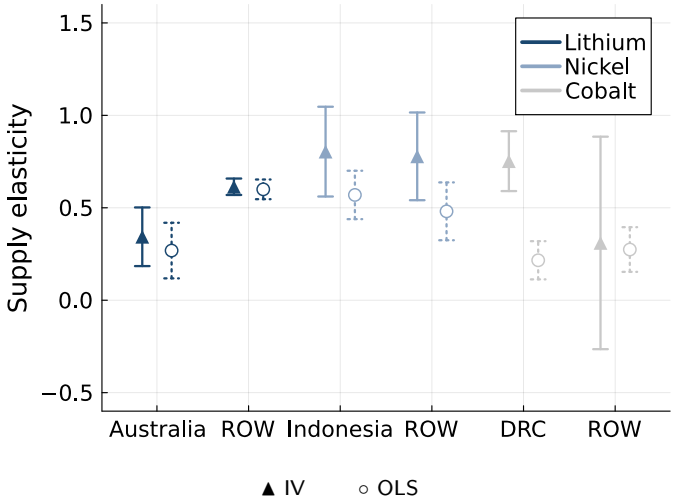
- To simplify, consider static FOC for estimation
  - Endogeneity problem from cost shocks $\varepsilon^{it}$
  - Need an instrument, which gives an additional moment

# Estimation with cost data

$$\underbrace{p^t g^i}_{mr^{it}} = \underbrace{\kappa x^{it} + \overbrace{\varepsilon^{it}}^{ac^{it} - \frac{\kappa}{2} x^{it}}}_{mc^{it}} + \eta^{it}$$

- We use an average cost moment
  - Marginal cost is $\kappa x + \varepsilon$, so average cost is $\frac{\kappa}{2} x + \varepsilon$
  - Because total cost is $(\frac{\kappa}{2} x + \varepsilon) x$
  - Invert $ac$ to isolate $\varepsilon$
- With dynamics, use lagged IV for expectational error

# Mineral elasticities




- Australian lithium is less elastic than other lithium producers
- Nickel is more elastic in general
- DRC cobalt is more elastic than other cobalt producers

# Demand

# Demand by battery $j$, region $k$, quarter $t$

$$w_{jkt} = \alpha_{jt} + \beta_j \log \frac{X_{kt}}{P_t} + \sum_{j'} \gamma_{jj'} \log p_{j't} + \varepsilon_{jkt}$$

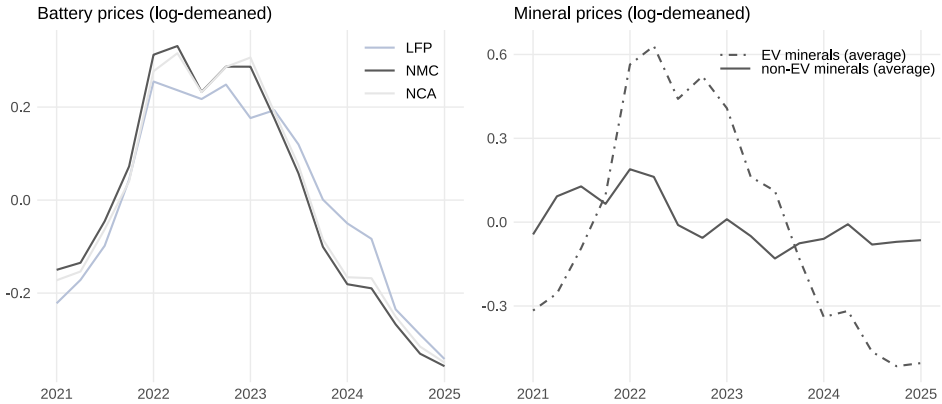
- Almost ideal demand system (Deaton & Muellbauer 1980)
  - Expenditure shares $w_{jkt}$, prices $p_{jt}$, translog price index $P_t$
- Avoids specifying the characteristics that govern substitution
- Endogeneity from demand shocks $\varepsilon_{jkt}$

# Estimation with instruments

$$z_{jt} = \sum_{m \in \mathcal{M}'} r_j^m p^{mt}, \quad \mathcal{M}' = \{\text{Al, Mn, Fe, Ph}\}$$

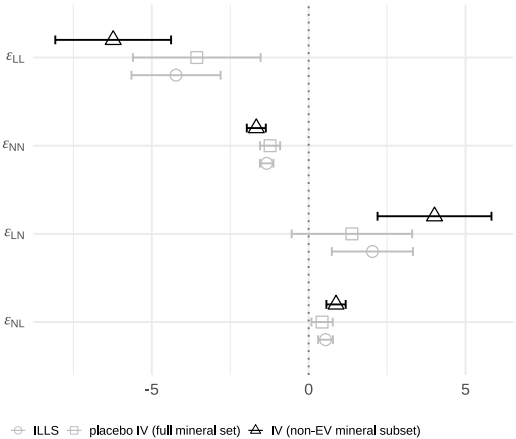
- Secondary mineral prices as cost shifters for battery prices
  - Relevance: minerals affect batteries through recipes $r_j^m$
  - Exclusion: EVs are less than $3\%$ of demand for $\mathcal{M}'$

# Price variation




- Recipe variation disentangles battery prices, which move together
  - Excluded: lithium, nickel, and cobalt, which move with batteries

# Battery elasticities




- IV $>$ OLS as expected
  - But only with $\mathcal{M}' \subset \mathcal{M}$
- $LL > NN$ reflects quality of $L$
  - Shorter driving range
  - More price-sensitive customers
- $LN$ and $NL$ capture switching

# Counterfactuals

# Equilibrium prices

- Mineral prices clear markets (with non-battery demand $\delta^{mt}$)

$$\sum_{i \in \mathcal{I}^{mt}} s^{it} = \sum_{j} r_j^m d_{jt} + \delta^{mt}$$

- Battery prices from mineral prices (with midstream wedge $\mu_{jt}$)

$$p_{jt} = \sum_{m} r_j^m p^{mt} + \mu_{jt}$$

- **Equilibrium**: sequence of prices $\{p^{mt}\}$ that clear markets $\forall\, m, t$

# Solving the model

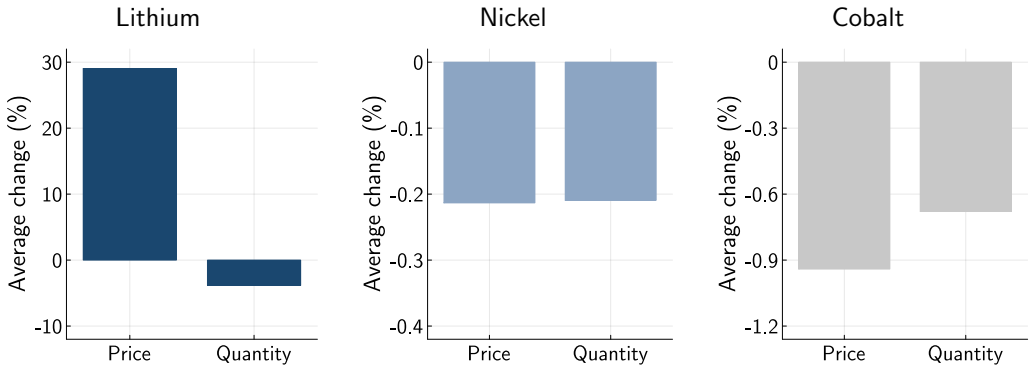
- A country restricts supply to exercise market power
  - Production tax maximizes producer surplus and government revenue
- Solve for equilibrium prices and production over time
  - Other producers respond to the new price path
  - Consumers substitute across battery technologies
- Compare to baseline equilibrium with no restrictive policies

# 1. Supply chain vulnerability

| | Price effects (%) | | | | |
|---|---|---|---|---|---|
| | Lithium | Nickel | Cobalt | Battery $L$ | Battery $N$ |
| Lithium (Australia) | 107 | −1.09 | −3.69 | 9.47 | 9.46 |
| Nickel (Indonesia) | −1.84 | 76.0 | −3.43 | −0.17 | 3.55 |
| Cobalt (DRC) | −6.19 | −3.74 | 865 | −0.58 | 14.1 |

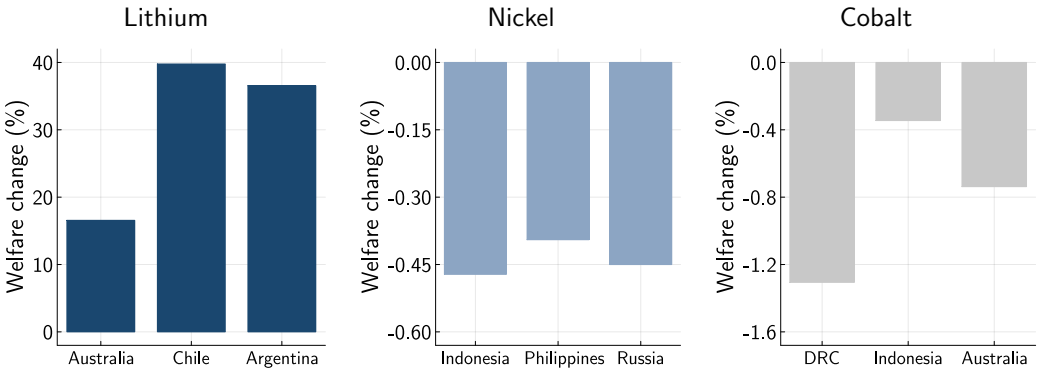
- Remove top producer, solve for equilibrium price changes
  - Large price increases (76% to 865%)
  - Lithium is common input (raises both battery prices)
  - Minerals are **gross complements** (negative cross-effects)

# 2. Unilateral policy: Australian lithium



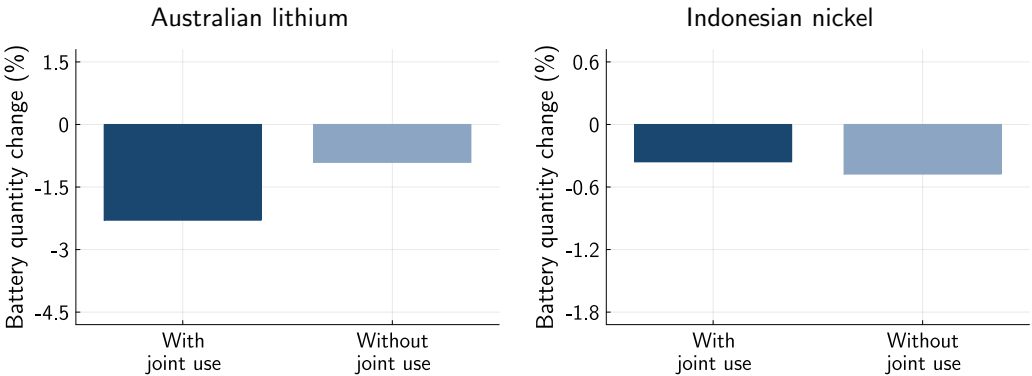

- Australia restricts lithium: lithium price rises, nickel and cobalt prices fall

# 2. Unilateral policy: Australian lithium and geopolitics



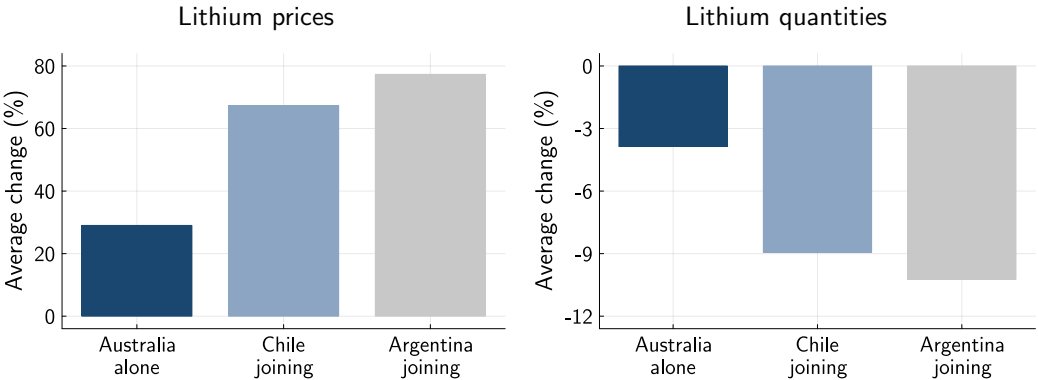

- Australia restricts lithium: substitutes win and free-ride, complements lose

# 2. Unilateral policy: green adoption and joint use



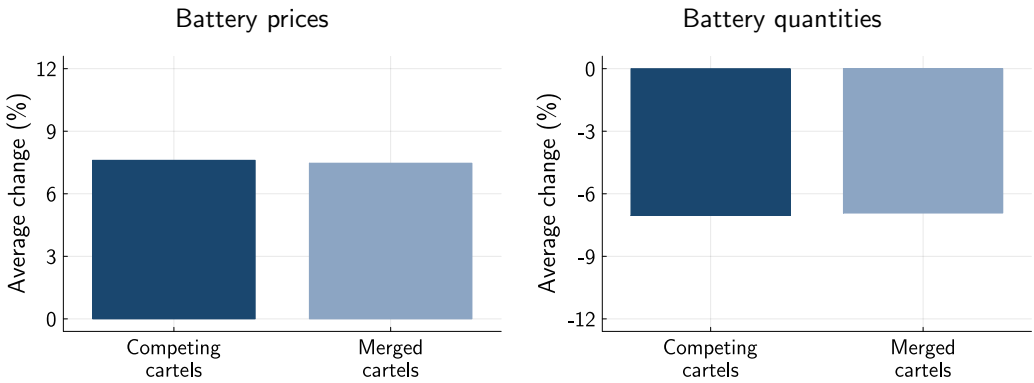

- Lithium is common input: joint use worsens losses (no substitute)
- Nickel is complement: joint use blunts losses ($L$ subsidy)

# 3. Cartel policy: OPEC for lithium



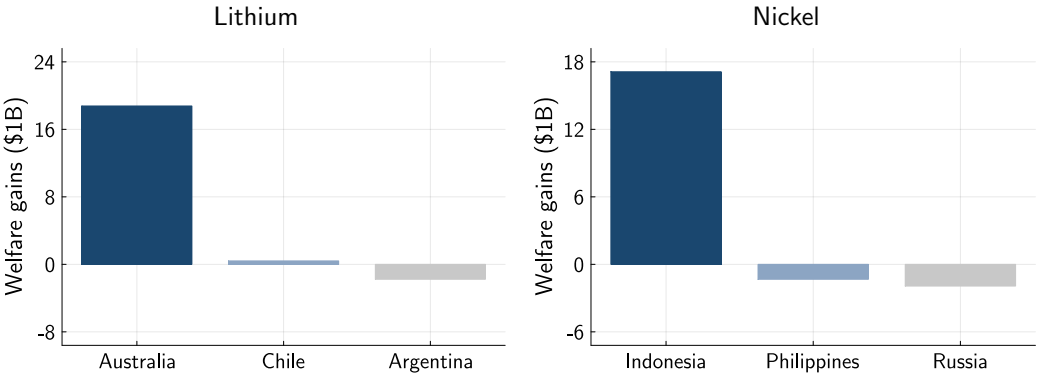

- Cartel restricts lithium: bigger cut, bigger effects

# 3. Cartel policy: OPEC for nickel and lithium



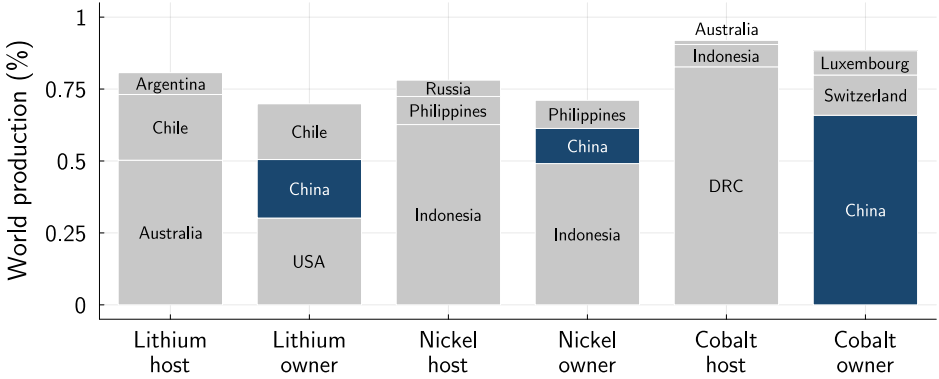

- Consolidation vs. complementarity (Cournot 1838)

# 3. Cartel policy: cohesion



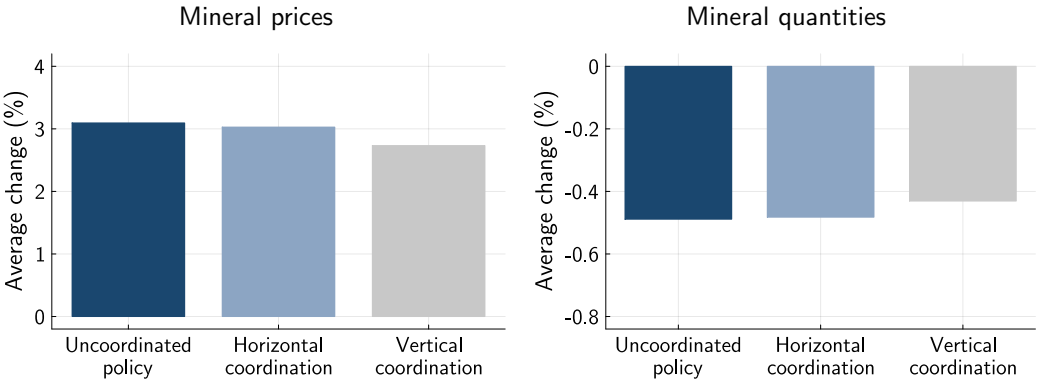

- Australia and Indonesia are anchors, but most others prefer to defect

# 4. Chinese policy: ownership



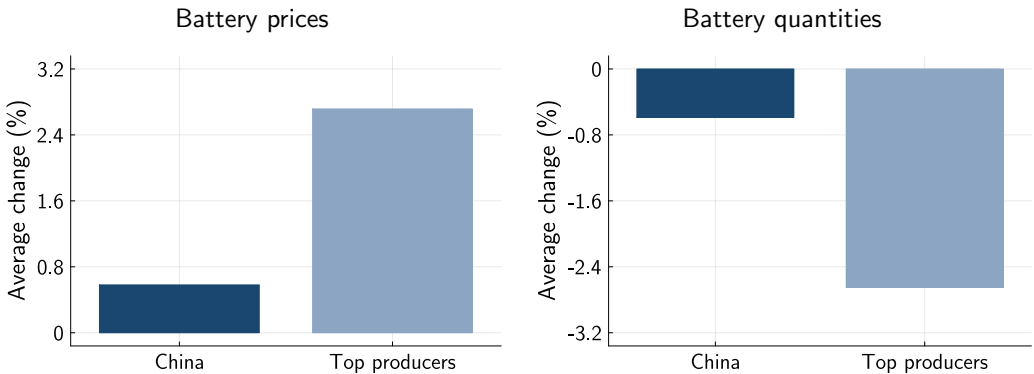

- Chinese companies own mines across minerals

# 4. Chinese policy: coordination




- Horizontal coordination internalizes cross-mineral complementarity
- Vertical coordination accounts for downstream battery consumption

# 5. Chinese policy: resource nationalism




- Chinese multinationals vs. independent Australian and Indonesian policy
  - Top producers have larger market shares and do not coordinate across minerals

# Summary

- Critical minerals are the next generation of energy resources
- These markets are **concentrated** and **interdependent**
  - With implications for geopolitics and green adoption